

WITTMUNDHAFEN, Germany

WMO: 101260

Lat: 53.5503N

Lon: 7.6672E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	21.5	13.2	10.8	20.5	17.1	13.1	23.2	32.6	45.4	29.4	45.5	8.5	100	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	84.4	67.6	80.2	66.4	76.0	64.4	69.4	81.0	67.4	77.2	65.6	74.2	9.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	94.3	73.5	63.9	89.1	71.9	62.3	84.2	70.3	33.4	81.5	31.8	77.2	30.5	74.3	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.8	19.1	13.2	91.2	5.4	2.5	9.4	93.0	6.2	94.5	3.2	95.9	-0.7	97.8		
(5)			12.9	73.3	5.0	1.4	9.3	74.3	6.3	75.1	3.5	76.0	-0.2	77.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	36.2	37.1	41.2	48.1	54.4	59.7	64.4	64.2	58.4	51.0	43.1	37.9		
	DBStd	11.87	7.50	7.00	6.07	6.51	6.22	5.65	5.83	5.24	4.74	5.82	6.03	7.44		
	HDD50	1859	429	362	279	112	24	0	0	0	1	58	216	378		
	HDD65	5738	894	781	739	507	334	181	84	79	207	434	656	842		
	CDD50	1753	0	1	5	56	161	290	448	439	252	89	10	2		
	CDD65	155	0	0	0	0	5	21	67	53	9	0	0	0		
	CDH74	1315	0	0	0	10	59	206	557	417	64	2	0	0		
	CDH80	377	0	0	0	0	8	56	175	131	7	0	0	0		
(14) Wind	WSAvg	9.1	10.9	10.6	10.5	9.1	8.7	8.3	7.8	7.4	7.8	8.8	9.2	10.1		
(15) Precipitation	PrecAvg	32.5	2.6	2.1	1.8	1.7	2.1	2.7	3.6	3.5	3.4	3.1	3.0	3.0		
	PrecMax	45.0	5.5	5.2	3.9	4.6	4.6	6.1	6.3	7.4	8.7	8.1	6.8	7.8		
	PrecMin	21.3	0.0	0.8	0.2	0.1	0.4	0.3	1.5	1.0	0.7	0.7	0.2	0.8		
	PrecStd	5.2	1.4	1.0	1.1	1.0	0.9	1.4	1.4	1.4	1.9	1.5	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	55.4	62.6	75.0	80.4	86.6	90.2	89.3	80.6	70.0	59.4	55.1		
		MCWB	50.4	49.0	51.4	58.7	64.0	68.9	69.6	68.6	67.4	61.7	55.7	52.4		
	2%	DB	50.2	51.3	57.0	68.9	74.8	80.2	84.9	83.8	75.1	64.6	56.4	52.1		
		MCWB	47.8	47.4	49.7	55.4	60.7	65.5	68.2	67.4	64.5	59.1	53.6	49.7		
	5%	DB	48.0	48.8	53.2	64.2	70.1	74.7	80.3	78.4	70.0	61.5	53.8	49.8		
		MCWB	46.1	46.0	47.8	54.1	59.0	63.5	67.0	66.1	61.6	57.2	51.1	48.1		
	10%	DB	46.0	46.6	50.2	59.3	66.2	70.1	75.6	73.9	66.5	59.2	51.6	47.6		
		MCWB	44.3	44.1	46.4	51.6	57.3	61.1	65.1	64.1	59.7	55.7	49.1	45.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	50.7	53.9	60.1	66.7	70.2	72.2	71.7	68.2	62.4	56.7	52.9		
		MCDB	52.8	53.1	59.1	71.3	76.4	83.5	84.8	83.0	78.1	68.6	59.1	54.6		
	2%	WB	48.3	48.0	50.9	57.0	62.6	67.0	69.7	68.8	65.2	59.7	54.1	50.4		
		MCDB	49.6	50.2	55.0	66.3	71.7	77.4	81.9	79.2	73.7	63.5	56.3	52.1		
	5%	WB	46.1	46.1	48.9	54.7	59.9	64.3	67.7	66.8	62.8	58.0	51.7	48.0		
		MCDB	47.5	48.5	52.7	62.6	67.8	72.3	77.7	76.1	68.4	61.1	53.8	49.4		
	10%	WB	44.0	44.1	46.5	52.4	57.6	62.1	65.8	65.1	60.8	56.3	49.5	45.7		
		MCDB	45.5	46.2	49.8	58.9	64.9	68.8	74.1	73.0	65.2	59.1	51.1	47.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.8	12.6	16.3	16.3	16.2	16.4	16.6	14.6	12.4	9.2	8.1		
		MCDBR	8.9	12.0	18.6	23.9	23.4	24.1	24.0	23.6	20.4	14.9	10.2	8.6		
		MCWBR	8.0	9.2	12.3	13.2	12.4	12.6	10.8	10.7	11.1	10.0	8.4	8.0		
	5% WB	MCDBR	8.5	10.2	16.4	21.3	20.8	21.4	21.6	20.4	17.8	13.8	9.8	8.4		
		MCWBR	8.1	8.6	11.6	12.7	12.0	12.2	10.6	10.4	10.6	9.8	8.6	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.346	0.375	0.407	0.402	0.401	0.415	0.410	0.387	0.362	0.341	0.320		
	taud		2.314	2.329	2.272	2.238	2.292	2.320	2.295	2.322	2.375	2.402	2.347	2.305		
	Ebn at Noon		200	236	253	260	268	270	263	257	248	228	192	177		
	Edh at Noon		19	26	33	39	39	38	38	36	30	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		181	388	779	1301	1623	1702	1612	1337	947	525	223	138		
	RadStd		18	58	81	131	141	125	189	106	96	55	21	15		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.85	+2.48	+2.77	N/A	N/A	N/A	N/A	-405	+112	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-236	N/A	N/A	

Nomenclature: See separate page